

Project: RetailCart ML Engine

Platform Name for Hugging-face: RetailCart

Transforming customer transactional data into actionable retention strategies using machine learning

Problem Statement:

RetailCart is a global e-commerce business operating across multiple markets, maintaining a transactional dataset of **2,240 active customers** tracked across **22 behavioral and demographic attributes**—including spending habits, channel engagement, demographic profiles, and historical campaign responsiveness.

Currently, RetailCart relies on **one-size-fits-all marketing strategies**. Treating a highly diverse customer base with blanket campaigns creates three critical business bottlenecks:

1. **Capital Inefficiency:** High marketing spend wasted on low-converting audiences.
2. **Revenue Leakage:** Failure to identify and offer premium incentives to high-value VIP buyers.
3. **Customer Churn:** Inability to proactively flag disengaged or churn-prone shoppers before they leave the platform.

The Objective:

To overcome these challenges, “RetailCart” requires a data-driven Customer Intelligence & Segmentation Engine powered by unsupervised machine learning.

By processing historical interaction and transaction metrics, the system will discover latent behavioral personas, partition users into distinct clusters based on loyalty and purchasing dynamics, and provide actionable insights to drive personalized marketing and retention workflows.

Solution roadmap:

1. Data Cleaning & Feature Engineering

- **Preprocessing:** Handle missing values (e.g., missing income data), fix data types, and handle potential outliers.
- **Feature Creation:** Derive high-signal customer metrics from the 22 raw attributes:
 - **Customer Lifetime / Tenure:** Calculated from enrollment date (**Dt_Customer**).
 - **Total Spend:** Aggregate spending across categories (Wines, Fruits, Meat, Fish, Sweets, Gold).
 - **Total Purchases:** Aggregate channel activity (Web, Catalog, Store purchases).
 - **Campaign Engagement:** Combined response rate across all campaign offers.
 - **Family Dynamics:** Calculate total household size from education, marital status, children, and teenagers.

2. Feature Scaling & Dimensionality Reduction

- **Scaling:** Apply **StandardScaler** or **RobustScaler** since clustering algorithms relying on distance metrics (like K-Means) are sensitive to feature scales.
- **Dimensionality Reduction:** Use **PCA (Principal Component Analysis)** or **t-SNE/UMAP** to remove multicollinearity, reduce noise, and enable visual interpretation of the clusters.

3. Clustering Models & Evaluation

- **Algorithms to Explore:**
 - **K-Means:** For clean, spherical clusters.
 - **Agglomerative Hierarchical Clustering:** To inspect natural hierarchy via dendrograms.
 - **DBSCAN / HDBSCAN:** To handle potential density-based groupings and identify extreme outlier profiles.
- **Optimal Cluster Determination:** Evaluate results using the **Elbow Method (Inertia)**, **Silhouette Score**, and **Davies-Bouldin Index**.

4. Cluster Profiling & Actionable Marketing Insights

Once clusters are formed, we translate mathematical groups into business personas:

- **High-Value Loyalists:** High spend, high engagement $\rightarrow$ *Target with VIP perks and early product access.*
- **Budget / Frequent Shoppers:** High transaction count, lower basket size $\rightarrow$ *Target with bundle discounts and cross-selling.*
- **At-Risk / Churn-Prone Customers:** Low recent activity, declining visits $\rightarrow$ *Target with re-engagement campaigns and personalized discounts.*
- **Discount Seekers:** High deal purchases $\rightarrow$ *Target with seasonal promotion alerts.*

